Inverse z-transform

Answers and worked solutions · Day 15 · Sat 7 Nov · pack D8

5. Practice

From your course files

T7 Q1 (tutorial).

Hint: factor the 3 out of $(3 - z^{-1})$ first, so that both factors read $(1 - pz^{-1})$.

$$X(z) = \frac{\frac{5}{3}z^{-1}}{(1 - 2z^{-1})\left(1 - \frac{1}{3}z^{-1}\right)} = \frac{1}{1 - 2z^{-1}} - \frac{1}{1 - \frac{1}{3}z^{-1}} \qquad \text{[partial fraction expansion]}$$

Two poles at radii $\frac{1}{3}$ and 2 give three possible ROCs, each with its own signal:

$$x(n) = \begin{cases} 2^n u(n) - (\frac{1}{3})^n u(n), & |z| > 2 \text{ (causal, unstable)} \\ -2^n u(-n-1) - (\frac{1}{3})^n u(n), & \frac{1}{3} < |z| < 2 \text{ (two-sided, the absolutely summable one)} \\ -2^n u(-n-1) + (\frac{1}{3})^n u(-n-1), & |z| < \frac{1}{3} \text{ (anti-causal)} \end{cases}$$

The official solution also lists $2^n u(n) + (\frac{1}{3})^n u(-n-1)$ with ROC $\varnothing$. That one is not a valid signal for this $X(z)$ (its two ROCs $|z| > 2$ and $|z| < \frac{1}{3}$ do not overlap), so there are 3 real answers.

T7 Q2 (tutorial).

Hint: $1 - \frac{10}{3}z^{-1} + z^{-2} = (1 - \frac{1}{3}z^{-1})(1 - 3z^{-1})$.

$$X(z) = \frac{3}{(1 - \frac{1}{3}z^{-1})(1 - 3z^{-1})} = -\frac{\frac{3}{8}}{1 - \frac{1}{3}z^{-1}} + \frac{\frac{27}{8}}{1 - 3z^{-1}} \qquad \text{[partial fraction expansion]}$$

Absolutely summable $x(n)$ means the ROC of $X(z)$ must include the unit circle $|z| = 1$, so the ROC is $\frac{1}{3} < |z| < 3$. The pole at $\frac{1}{3}$ gives a causal term and the pole at 3 an anti-causal term:

$$x(n) = -\frac{3}{8}\left(\frac{1}{3}\right)^n u(n) - \frac{27}{8}\,3^n u(-n-1), \qquad \text{ROC: } \frac{1}{3} < |z| < 3.$$

T7 Q3 (tutorial).

Hint: combine $X(z)$ into one fraction first; then $H = Y/X$.

(a) Using the delay property on the second term of $x(n)$:

$$X(z) = \frac{1}{1-\frac{1}{2}z^{-1}} - \frac{\frac{1}{4}z^{-1}}{1-\frac{1}{2}z^{-1}} = \frac{1-\frac{1}{4}z^{-1}}{1-\frac{1}{2}z^{-1}}, \qquad Y(z) = \frac{1}{1-\frac{1}{3}z^{-1}}.$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1-\frac{1}{2}z^{-1}}{\left(1-\frac{1}{4}z^{-1}\right)\left(1-\frac{1}{3}z^{-1}\right)} = \frac{1-\frac{1}{2}z^{-1}}{1-\frac{7}{12}z^{-1}+\frac{1}{12}z^{-2}} = \frac{3}{1-\frac{1}{4}z^{-1}} - \frac{2}{1-\frac{1}{3}z^{-1}}.$$

As the system is causal, the ROC is $|z| > \frac{1}{3}$ and

$$h(n) = 3\left(\frac{1}{4}\right)^n u(n) - 2\left(\frac{1}{3}\right)^n u(n).$$

(b) Cross-multiply $Y(z)\left(1-\frac{7}{12}z^{-1}+\frac{1}{12}z^{-2}\right) = X(z)\left(1-\frac{1}{2}z^{-1}\right)$ and invert term by term:

$$y(n) - \frac{7}{12}y(n-1) + \frac{1}{12}y(n-2) = x(n) - \frac{1}{2}x(n-1).$$

(c) Stable, because the ROC of $H(z)$, $|z| > \frac{1}{3}$, includes the unit circle $|z| = 1$.

All three answers were reproduced in Python (sympy partial fractions) and agree with the official solutions.

T8 Q5(c) (tutorial, optional redo).

The full worked solution is Example 2 in section 3 above (and parts (a)–(b) are in pack D10). Key steps:

The zeros of H are 0.8 and $-\frac{3}{2}$ (one inside, one outside the unit circle), so the channel is nonminimum-phase and its stable inverse is noncausal:

$$H_I(z) = H^{-1}(z) = \frac{1-0.8z^{-1}+0.64z^{-2}}{1+0.7z^{-1}-1.2z^{-2}}, \qquad \text{ROC: } 0.8 < |z| < \tfrac{3}{2}.$$

Partial fractions (equal degrees, so a constant $0.64/(-1.2) = -0.5333$ appears):

$$H_I(z) = \frac{0.3478}{1-0.8z^{-1}} + \frac{1.1855}{1+\frac{3}{2}z^{-1}} - 0.5333$$

$$h_I(n) = 0.3478\,(0.8)^n u(n) - 1.1855\left(-\tfrac{3}{2}\right)^n u(-n-1) - 0.5333\,\delta(n),$$

which is absolutely summable, as expected. **Noncausal** (two-sided).

New questions

Q1. Power series [4 marks]

a. Coefficient of z^{-n} is $x(n)$: $z^2 \to x(-2) = 2$, $z^1 \to x(-1) = -1$, $z^0 \to x(0) = 3$, $z^{-2} \to x(2) = -4$.

$$x(n) = \{2,\ -1,\ \underset{\uparrow}{3},\ 0,\ -4\}$$

b. Divide $1 + 2z^{-1}$ by $1 - z^{-1} + \frac{1}{2}z^{-2}$ in ascending powers of z^{-1}:

- 1: remainder $(1 + 2z^{-1}) - (1 - z^{-1} + \frac{1}{2}z^{-2}) = 3z^{-1} - \frac{1}{2}z^{-2}$.
- $3z^{-1}$: remainder $(3z^{-1} - \frac{1}{2}z^{-2}) - (3z^{-1} - 3z^{-2} + \frac{3}{2}z^{-3}) = \frac{5}{2}z^{-2} - \frac{3}{2}z^{-3}$.
- $\frac{5}{2}z^{-2}$: remainder $(\frac{5}{2}z^{-2} - \frac{3}{2}z^{-3}) - (\frac{5}{2}z^{-2} - \frac{5}{2}z^{-3} + \frac{5}{4}z^{-4}) = z^{-3} - \frac{5}{4}z^{-4}$.
- z^{-3}.

$$x(0) = 1,\ x(1) = 3,\ x(2) = 2.5,\ x(3) = 1$$

c. The poles are $\frac{1}{2} \pm j\frac{1}{2}$, radius $1/\sqrt{2}$, so this ROC is the left-sided one. Divide in ascending powers of z: write $\dfrac{2z^{-1} + 1}{\frac{1}{2}z^{-2} - z^{-1} + 1}$. First term $2z^{-1}/(\frac{1}{2}z^{-2}) = 4z$.

Remainder
$(2z^{-1} + 1) - 4z(\frac{1}{2}z^{-2} - z^{-1} + 1) = (2z^{-1} + 1) - (2z^{-1} - 4 + 4z) = 5 - 4z$.
Next term $5/(\frac{1}{2}z^{-2}) = 10z^2$.

Series $4z + 10z^2 + \cdots$, so $x(-1) = 4$, $x(-2) = 10$ (and $x(n) = 0$ for $n \geq 0$). Checked in Python: the next terms are $x(-3) = 12$, $x(-4) = 4$.

Q2. LCCDE, improper $H(z)$ and system types [7 marks]

a. $Y(z)(1 - 0.5z^{-1}) = X(z)(1 + z^{-1})$, so

$$H_1(z) = \frac{1 + z^{-1}}{1 - 0.5z^{-1}} = \frac{z + 1}{z - 0.5}$$

Zero at $z = -1$, pole at $z = 0.5$. ROC $|z| > 0.5$ (causal).

b. $M = N = 1$, so divide first. Quotient $= 1/(-0.5) = -2$. Remainder:
$(1 + z^{-1}) - (-2)(1 - 0.5z^{-1}) = 3$.

$$H_1(z) = -2 + \frac{3}{1 - 0.5z^{-1}} \quad \Rightarrow \quad \mathbf{h_1(n) = -2\delta(n) + 3(0.5)^{nu}(n)}$$

Check: $h_1(0) = 1$, $h_1(1) = 1.5$, $h_1(2) = 0.75$, the same as running the recursion.

c.

System	$H(z)$	Type	Poles	Stable?
S1	$\dfrac{1+z^{-1}}{1-0.5z^{-1}}$	pole-zero (ARMA)	0.5	yes, $0.5<1$
S2	$1-\frac{1}{8}z^{-3}=\dfrac{z^3-\frac{1}{8}}{z^3}$	FIR (MA)	triple pole at $z=0$; zeros at $0.5e^{j2\pi k/3}$, $k=0,1,2$	yes, always (FIR)
S3	$\dfrac{1}{1-0.9z^{-2}}=\dfrac{z^2}{z^2-0.9}$	all-pole (AR)	$\pm\sqrt{0.9}=\pm0.9487$; double zero at $z=0$	yes, $0.9487<1$

Q3. Find all possible sequences [7 marks]

a. Numerator degree 2 < denominator degree 3: proper. Cover-up with $w=z^{-1}$:

$$A_{1/2}=\left[\frac{1+w^2}{(1-\frac{1}{3}w)(1-3w)}\right]_{w=2}=\frac{5}{(\frac{1}{3})(-5)}=-3$$

$$A_{1/3}=\left[\frac{1+w^2}{(1-\frac{1}{2}w)(1-3w)}\right]_{w=3}=\frac{10}{(-\frac{1}{2})(-8)}=\frac{5}{2}$$

$$A_{3}=\left[\frac{1+w^2}{(1-\frac{1}{2}w)(1-\frac{1}{3}w)}\right]_{w=1/3}=\frac{10/9}{(\frac{5}{6})(\frac{8}{9})}=\frac{3}{2}$$

$$X(z)=\frac{-3}{1-\frac{1}{2}z^{-1}}+\frac{5/2}{1-\frac{1}{3}z^{-1}}+\frac{3/2}{1-3z^{-1}}$$

Check at $w=0$: $-3+2.5+1.5=1=X|_{w=0}$.

b. Pole radii $\frac{1}{3}<\frac{1}{2}<3$ give 4 ROCs:

ROC	$x(n)$	Type
$\lvert z\rvert > 3$	$\left[-3(\frac{1}{2})^n + \frac{5}{2}(\frac{1}{3})^n + \frac{3}{2}\,3^n\right]u(n)$	causal (unstable)
$\frac{1}{2} < \lvert z\rvert < 3$	$\left[-3(\frac{1}{2})^n + \frac{5}{2}(\frac{1}{3})^n\right]u(n) - \frac{3}{2}\,3^n u(-n-1)$	two-sided (stable)
$\frac{1}{3} < \lvert z\rvert < \frac{1}{2}$	$\frac{5}{2}(\frac{1}{3})^n u(n) + \left[3(\frac{1}{2})^n - \frac{3}{2}\,3^n\right]u(-n-1)$	two-sided (unstable)
$\lvert z\rvert < \frac{1}{3}$	$\left[3(\frac{1}{2})^n - \frac{5}{2}(\frac{1}{3})^n - \frac{3}{2}\,3^n\right]u(-n-1)$	anti-causal (unstable)

The other $2^3 - 4 = 4$ sign combinations have an empty ROC.

c. Only $\frac{1}{2} < \lvert z\rvert < 3$ contains $\lvert z\rvert = 1$, so only that $x(n)$ has a DTFT.

Check: $X(1) = \dfrac{2}{(\frac{1}{2})(\frac{2}{3})(-2)} = -3$. From the sequence:
$\sum_n x(n) = -3 \cdot 2 + \frac{5}{2} \cdot \frac{3}{2} - \frac{3}{2}\sum_{n\le -1} 3^n = -6 + 3.75 - \frac{3}{2} \cdot \frac{1}{2} = -3$. They agree.

Q4. System ID → LCCDE → stability [10 marks]

a. $X(z) = \dfrac{1}{1 - z^{-1}}$, $\lvert z\rvert > 1$.

$$Y(z) = \frac{2/3}{1 - z^{-1}} - \frac{1/5}{1 - \frac{1}{2}z^{-1}} + \frac{8/15}{1 + 2z^{-1}} = \frac{1}{(1 - z^{-1})(1 - \frac{1}{2}z^{-1})(1 + 2z^{-1})}, \quad \lvert z\rvert > 2$$

(Combine over the common denominator. Numerator check: at $w = 0$, $\frac{2}{3} - \frac{1}{5} + \frac{8}{15} = 1$; the w and w^2 coefficients cancel, which Python confirms.)

b.

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 + 2z^{-1})} = \frac{1}{1 + \frac{3}{2}z^{-1} - z^{-2}}$$

Cross-multiply: $Y(z)(1 + \frac{3}{2}z^{-1} - z^{-2}) = X(z)$, so

$$\mathbf{y(n) + \tfrac{3}{2}y(n-1) - y(n-2) = x(n)}$$

It is an all-pole system, with poles 0.5 and -2 and a double zero at $z = 0$.

c. $H(z) = \dfrac{1/5}{1 - \frac{1}{2}z^{-1}} + \dfrac{4/5}{1 + 2z^{-1}}$ (cover-up: $w = 2 \Rightarrow 1/(1+4) = \frac{1}{5}$;
$w = -\frac{1}{2} \Rightarrow 1/(1 + \frac{1}{4}) = \frac{4}{5}$). Causal → ROC $\lvert z\rvert > 2$:

$$\mathbf{h(n) = \left[\frac{1}{5}\left(\frac{1}{2}\right)^{n} + \frac{4}{5}(-2)^{n}\right]u(n)}$$

Unstable. The ROC $|z| > 2$ excludes $|z| = 1$ (the pole at -2 is outside the unit circle). Consistency check: $R_h \cap R_x = \{|z| > 2\}$, which matches the ROC of Y.

d. Stable → the ROC must contain $|z| = 1$ → $\frac{1}{2} < |z| < 2$. Pole $\frac{1}{2}$ is causal, pole -2 is anti-causal:

$$\mathbf{h(n) = \frac{1}{5}\left(\frac{1}{2}\right)^{nu}(n) - \frac{4}{5}(-2)^{nu}(-n-1)}$$

Not causal (two-sided): $h(n) \neq 0$ for $n < 0$.

e. No. Now $R_y = R_h \cap R_x = \{1 < |z| < 2\}$, so the $(-2)^n$ term in the output becomes anti-causal: $y(n) = \left[\frac{2}{3} - \frac{1}{5}\left(\frac{1}{2}\right)^n\right]u(n) - \frac{8}{15}(-2)^n u(-n-1)$. Same LCCDE, different ROC, different system.

All values were checked in Python (scipy `lfilter` reproduces $y(0..5) = 1, -0.5, 2.75, -3.625, 9.1875, -16.406$ from the causal h).

Q5. Improper $H(z)$ with complex poles [8 marks]

a. Zeros: $1 + z^{-2} = 0 \Rightarrow z = \pm j$ (on the unit circle). Poles: $p^2 - p + \frac{1}{2} = 0 \Rightarrow p = \frac{1}{2} \pm j\frac{1}{2} = 0.7071e^{\pm j\pi/4}$. Causal, and both poles are inside the unit circle, so it is **stable**.

b. Divide in ascending powers of z^{-1}:

- 1: remainder $(1 + z^{-2}) - (1 - z^{-1} + \frac{1}{2}z^{-2}) = z^{-1} + \frac{1}{2}z^{-2}$
- z^{-1}: remainder $(z^{-1} + \frac{1}{2}z^{-2}) - (z^{-1} - z^{-2} + \frac{1}{2}z^{-3}) = \frac{3}{2}z^{-2} - \frac{1}{2}z^{-3}$
- $\frac{3}{2}z^{-2}$: remainder $(\frac{3}{2}z^{-2} - \frac{1}{2}z^{-3}) - (\frac{3}{2}z^{-2} - \frac{3}{2}z^{-3} + \frac{3}{4}z^{-4}) = z^{-3} - \frac{3}{4}z^{-4}$
- z^{-3}

$$h(0) = 1,\ h(1) = 1,\ h(2) = 1.5,\ h(3) = 1$$

c. $M = N = 2$: quotient $= 1/\frac{1}{2} = 2$. Remainder $(1 + z^{-2}) - 2(1 - z^{-1} + \frac{1}{2}z^{-2}) = -1 + 2z^{-1}$:

$$H(z) = 2 + \frac{-1 + 2z^{-1}}{1 - z^{-1} + \frac{1}{2}z^{-2}}$$

Cover-up for $p = \frac{1}{2} + j\frac{1}{2}$: $1/p = 1 - j$. Numerator: $-1 + 2(1 - j) = 1 - 2j$. Other factor: $1 - p^*(1/p) = 1 + j$ (as in Example 3b).

$$A = \frac{1 - 2j}{1 + j} = \frac{(1 - 2j)(1 - j)}{2} = -\frac{1}{2} - j\frac{3}{2}, \qquad |A| = \frac{\sqrt{10}}{2},\ \angle A = -108.43^\circ$$

$2r^n[\mathrm{Re}\, A \cos\theta n - \mathrm{Im}\, A \sin\theta n] = r^n[-\cos\frac{\pi n}{4} + 3\sin\frac{\pi n}{4}]$, so

$$\boxed{h(n) = 2\delta(n) + \left(\frac{1}{\sqrt{2}}\right)^n \left[3\sin\frac{\pi n}{4} - \cos\frac{\pi n}{4}\right] u(n)}$$

Equivalently $2\delta(n) + \sqrt{10}\,(0.7071)^n \cos(\frac{\pi n}{4} - 108.43^\circ)u(n)$.

Check: $n = 0$: $2 + (0 - 1) = 1$. $n = 1$: $0.7071(2.1213 - 0.7071) = 1$. $n = 2$: $0.5(3 - 0) = 1.5$. $n = 3$: $0.3536(2.1213 + 0.7071) = 1$. Matches (b). Python's `lfilter` agrees to $n = 9$.

d. A zero at $z = e^{j\pi/2} = j$ means $H(\omega)|_{\omega=\pi/2} = 0$. The steady-state output is **0**.

6. Quick recall check

Question

1. $-4(0.8)^n u(-n-1)$.
2. No. Causal means ROC $|z| > 1.1$, which excludes the unit circle.
3. Four (ROCs: $|z| < 0.5$, $0.5 < |z| < 2$, $2 < |z| < 5$, $|z| > 5$). The stable one is $0.5 < |z| < 2$.
4. $y(n) - 0.25y(n-2) = x(n) + 2x(n-1)$.
5. FIR (all-zero, MA). Triple pole at $z = 0$. Always stable.